

Use-Case Flow Specification

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #44 – Database Query Performance Profiler

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Use Case: Generate Weekly Performance Digest (UC-04)

Use Case ID	UC-04	Name	Generate Weekly Performance Digest
Actors	Database Administrator, Backend Lead (receive report); Scheduled Job Trigger (initiates)	Related FRs	FR-004, FR-002, FR-003
Trigger	Cron job fires every Monday at 08:00 AM		

Preconditions:

- At least 7 days of slow query log data has been ingested into the system.
- The Monday cron job is set up and running.
- At least one DBA or Backend Lead email address is configured.

Postconditions:

- A PDF digest has been generated and emailed to all configured recipients.
- The event is recorded in the audit log.
- Dashboard shows updated 'Last Report Sent' timestamp.

Main Success Scenario:

1. Cron job fires at 8:00 AM on Monday and triggers the digest service.
2. System pulls all slow query records from the past 7 days. Also fetches the same window from the previous week for comparison.
3. Queries are ranked by total execution time. Top 10 slowest are selected. For each one, the index recommendation engine runs and attaches suggestions.
4. System calculates week-over-week change in average query latency and total slow query count. Labels the trend as Improved, Same or Worsened.
5. System generates a PDF with three sections: (i) summary with trend label, (ii) top 10 slow queries with recommendations, (iii) list of suggested indexes.
6. PDF is emailed to all DBA and Backend Lead recipients. Subject line: '[DB Profiler] Weekly Digest – [date]'.
7. On delivery confirmation, the system logs the event and updates the 'Last Report Sent' timestamp on the dashboard.
8. Use case ends.

Alternate Flow 1 – Not enough data (Step 2):

- System finds fewer than 100 log entries for the week (e.g. the agent was down).
- Instead of a full report, system sends a short email saying there wasn't enough data and asking the DBA to check if the log agent is running.
- Logs a warning. Skips report for this week. Will try again next Monday.

Alternate Flow 2 – Email fails to send (Step 6):

- Email server returns an error or times out.
- System retries up to 3 times (waits 5 min, then 15 min, then 30 min between attempts).
- If all retries fail, the PDF is saved to a 'Failed Deliveries' folder. DBA gets an in-app notification to download it manually. Error is logged.